

# SAFETY DATA SHEET

Revision date 03-Sep-2024

Revision Number 1

## Section 1: Identification

### Product identifier

**Product Name** Immun-Star Substrate  
**Catalogue Number(s)** 1705018, 1705018EDU, 9702837, 9702838

### Other means of identification

### Recommended use of the chemical and restrictions on use

**Recommended use** Laboratory chemicals  
**Uses advised against** No information available

### Details of the supplier of the safety data sheet

| <u>Supplier</u>                                                                   | <u>Manufacturer</u>                                                                                      | <u>Importer</u>                                                                 |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Bio-Rad Laboratories Inc.<br>1000 Alfred Nobel Drive<br>Hercules, CA 94547<br>USA | Bio-Rad Laboratories, Life Science Group<br>2000 Alfred Nobel Drive<br>Hercules, California 94547<br>USA | Bio-Rad Laboratories Pty Ltd<br>189 Bush Road<br>Albany Auckland<br>New Zealand |
| <b>Technical Service</b>                                                          | +64 9 415 2280 or 0508 805 500<br>sales.nz@bio-rad.com                                                   |                                                                                 |

### Emergency telephone number

**24 Hour Emergency Phone Number** CHEMTREC New Zealand: 64-98010034

## Section 2: Hazard identification

### GHS Classification

|                                                    |            |
|----------------------------------------------------|------------|
| Serious eye damage/eye irritation                  | Category 2 |
| Carcinogenicity                                    | Category 2 |
| Specific target organ toxicity — single exposure   | Category 2 |
| Specific target organ toxicity — repeated exposure | Category 2 |

### Label elements



**Signal word**  
Warning

**Hazard statements**  
Causes serious eye irritation  
Suspected of causing cancer  
May cause damage to organs  
May cause damage to organs through prolonged or repeated exposure

**Precautionary Statements - Prevention**

Wear protective gloves/clothing and eye/face protection  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product

**Precautionary Statements - Response**

IF exposed or concerned: Call a POISON CENTER or doctor

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
If eye irritation persists: Get medical advice/attention

**Precautionary Statements - Disposal**

Dispose of contents/container in accordance with local, regional, national, and international regulations as applicable

**Other hazards which do not result in classification**

No information available.

**Section 3: Composition/information on ingredients**

| Chemical name  | CAS No.  | Weight-% |
|----------------|----------|----------|
| Diethanolamine | 111-42-2 | 1 - 2.5  |

|                           |             |         |
|---------------------------|-------------|---------|
| Non-hazardous ingredients | Proprietary | Balance |
|---------------------------|-------------|---------|

**Section 4: First-aid measures****Description of first aid measures**

|                                           |                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General advice</b>                     | IF exposed or concerned: Get medical advice/attention. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                         |
| <b>Inhalation</b>                         | Remove to fresh air. IF exposed or concerned: Get medical advice/attention.                                                                                                                                                                                                                                             |
| <b>Eye contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. If symptoms persist, call a doctor. Remove contact lenses, if present and easy to do. Continue rinsing. Keep eye wide open while rinsing. Do not rub affected area. Get medical attention if irritation develops and persists. |
| <b>Skin contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If symptoms persist, call a doctor.                                                                                                                                                                                                                  |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Rinse mouth. Never give anything by mouth to an unconscious person. Call a doctor.                                                                                                                                                                                                              |
| <b>Self-protection of the first aider</b> | Avoid contact with skin, eyes or clothing. Wear personal protective clothing (see section 8).                                                                                                                                                                                                                           |

**Most important symptoms and effects, both acute and delayed**

|                            |                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Symptoms</b>            | May cause redness and tearing of the eyes. Burning sensation. Prolonged contact may cause redness and irritation. |
| <b>Effects of Exposure</b> | May cause damage to organs. May cause damage to organs through prolonged or repeated exposure.                    |

**Indication of any immediate medical attention and special treatment needed**

**Note to doctors** Treat symptomatically.

## Section 5: Fire-fighting measures

### Suitable Extinguishing Media

**Suitable Extinguishing Media** Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Large Fire** CAUTION: Use of water spray when fighting fire may be inefficient.

**Unsuitable extinguishing media** Do not scatter spilled material with high pressure water streams.

### Specific hazards arising from the chemical

**Specific hazards arising from the chemical** No information available.

### Special protective actions for fire-fighters

**Special protective equipment and precautions for fire-fighters** Firefighters should wear self-contained breathing apparatus and full firefighting turnout gear.

## Section 6: Accidental release measures

### Personal precautions, protective equipment and emergency procedures

**Personal precautions** Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Avoid contact with skin, eyes or clothing.

**Other information** Refer to protective measures listed in Sections 7 and 8.

**For emergency responders** Use personal protection recommended in Section 8.

### Environmental precautions

**Environmental precautions** See Section 12 for additional Ecological Information.

### Methods and material for containment and cleaning up

**Methods for containment** Prevent further leakage or spillage if safe to do so.

**Methods for cleaning up** Pick up and transfer to properly labelled containers.

### Precautions to prevent secondary hazards

**Prevention of secondary hazards** Clean contaminated objects and areas thoroughly observing environmental regulations.

## Section 7: Handling and storage

### Precautions for safe handling

**Advice on safe handling** Handle in accordance with good industrial hygiene and safety practice. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product.

**General hygiene considerations** Do not eat, drink or smoke when using this product. Wash hands before breaks and immediately after handling the product. Avoid contact with skin, eyes or clothing. Wear suitable gloves and eye/face protection.

**Conditions for safe storage, including any incompatibilities**

**Storage Conditions** Store locked up. Keep containers tightly closed in a dry, cool and well-ventilated place. Store according to product and label instructions.

**Incompatible materials** None known based on information supplied.

## Section 8: Exposure controls/personal protection

**Control parameters****Exposure Limits**

| Chemical name              | New Zealand                                     | Australia                               | ACGIH TLV                                                         | United Kingdom |
|----------------------------|-------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------|----------------|
| Diethanolamine<br>111-42-2 | TWA: 3 ppm<br>TWA: 13 mg/m <sup>3</sup><br>Skin | TWA: 3 ppm<br>TWA: 13 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>inhalable fraction and<br>vapor<br>S* | -              |

**Biological occupational exposure limits** This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies.

**Appropriate engineering controls**

**Engineering controls** Showers  
Eyewash stations  
Ventilation systems.

**Individual protection measures, such as personal protective equipment**

**Eye/face protection** Wear safety glasses with side shields (or goggles).

**Hand protection** Wear suitable gloves.

**Skin and body protection** Wear suitable protective clothing.

**Respiratory protection** No protective equipment is needed under normal use conditions. If exposure limits are exceeded or irritation is experienced, ventilation and evacuation may be required.

**Environmental exposure controls** No information available.

## Section 9: Physical and chemical properties

**Information on basic physical and chemical properties**

**Physical state** Liquid  
**Appearance** aqueous solution  
**Colour** colourless  
**Odour** Odourless.  
**Odour threshold** No information available

| <u>Property</u>                         | <u>Values</u>     | <u>Remarks • Method</u> |
|-----------------------------------------|-------------------|-------------------------|
| pH                                      | 10.6              |                         |
| Melting point / freezing point          | 0 °C              |                         |
| Initial boiling point and boiling range | 100 °C            |                         |
| Flash point                             | No data available | None known              |
| Evaporation rate                        | No data available | None known              |

|                                        |                           |            |
|----------------------------------------|---------------------------|------------|
| Flammability                           | No data available         | None known |
| Flammability Limit in Air              |                           | None known |
| Upper flammability or explosive limits | No data available         |            |
| Lower flammability or explosive limits | No data available         |            |
| Vapour pressure                        | No data available         | None known |
| Relative vapour density                | No data available         | None known |
| Relative density                       | No data available         | None known |
| Water solubility                       | Miscible in water         |            |
| Solubility(ies)                        | No data available         | None known |
| Partition coefficient                  | No data available         | None known |
| Autoignition temperature               | No data available         | None known |
| Decomposition temperature              |                           | None known |
| Kinematic viscosity                    | No data available         | None known |
| Dynamic viscosity                      | No data available         | None known |
| Explosive properties                   | No information available. |            |
| Oxidising properties                   | No information available. |            |
| <b>Other information</b>               |                           |            |
| Softening point                        | No information available  |            |
| Molecular weight                       | No information available  |            |
| VOC content                            | No information available  |            |
| Liquid Density                         | No information available  |            |
| Bulk density                           | No information available  |            |
| Particle characteristics               | No information available  |            |

## Section 10: Stability and reactivity

### Reactivity

Reactivity No information available.

### Chemical stability

Stability Stable under normal conditions.

### Explosion data

Sensitivity to mechanical impact None.

Sensitivity to static discharge None.

### Possibility of hazardous reactions

Possibility of hazardous reactions None under normal processing.

### Conditions to avoid

Conditions to avoid None known based on information supplied.

### Incompatible materials

Incompatible materials None known based on information supplied.

### Hazardous decomposition products

Hazardous decomposition products None known based on information supplied.

## Section 11: Toxicological information

**Acute toxicity****Information on likely routes of exposure****Product Information**

|                     |                                                                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>   | Specific test data for the substance or mixture is not available. May cause irritation of respiratory tract.                                                 |
| <b>Eye contact</b>  | Specific test data for the substance or mixture is not available. Causes serious eye irritation (based on components). May cause redness, itching, and pain. |
| <b>Skin contact</b> | Specific test data for the substance or mixture is not available. Prolonged contact may cause redness and irritation. Causes mild skin irritation.           |
| <b>Ingestion</b>    | Specific test data for the substance or mixture is not available. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhoea.           |
| <b>Symptoms</b>     | May cause redness and tearing of the eyes. Prolonged contact may cause redness and irritation.                                                               |

**Acute toxicity****Numerical measures of toxicity**

The following values are calculated based on chapter 3.1 of the GHS document

ATEmix (oral) 33,790.00 mg/kg

**Component Information**

| Chemical name  | Oral LD50           | Dermal LD50             | Inhalation LC50 |
|----------------|---------------------|-------------------------|-----------------|
| Diethanolamine | = 780 mg/kg ( Rat ) | = 11.9 mL/kg ( Rabbit ) | -               |

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

|                                          |                                                                                                                                |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin corrosion/irritation</b>         | Classification based on data available for ingredients. Causes mild skin irritation.                                           |
| <b>Serious eye damage/eye irritation</b> | Classification based on data available for ingredients. Causes serious eye irritation.                                         |
| <b>Respiratory or skin sensitisation</b> | No information available.                                                                                                      |
| <b>Germ cell mutagenicity</b>            | No information available.                                                                                                      |
| <b>Carcinogenicity</b>                   | Contains a known or suspected carcinogen. Classification based on data available for ingredients. Suspected of causing cancer. |

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Chemical name             | New Zealand | IARC     |
|---------------------------|-------------|----------|
| Diethanolamine - 111-42-2 | -           | Group 2B |

**Legend**

IARC (International Agency for Research on Cancer)

Group 2B - Possibly Carcinogenic to Humans

|                              |                           |
|------------------------------|---------------------------|
| <b>Reproductive toxicity</b> | No information available. |
|------------------------------|---------------------------|

|                                                 |                                                                                                 |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>STOT - single exposure</b>                   | May cause damage to organs.                                                                     |
| <b>STOT - repeated exposure</b>                 | May cause damage to organs through prolonged or repeated exposure.                              |
| <b>Aspiration hazard</b>                        | No information available.                                                                       |
| <b>Data used to identify the health effects</b> | Refer to Section 16 for Key literature references and sources for data used to compile the SDS. |

## Section 12: Ecological information

### Ecotoxicity

**Aquatic ecotoxicity** The environmental impact of this product has not been fully investigated.

| Chemical name  | Algae/aquatic plants                                                                                                        | Fish                                                                                                                                                                            | Crustacea                                  |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Diethanolamine | EC50: =7.8mg/L (72h, <i>Desmodesmus subspicatus</i> )<br>EC50: 2.1 - 2.3mg/L (96h, <i>Pseudokirchneriella subcapitata</i> ) | LC50: 4460 - 4980mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 1200 - 1580mg/L (96h, <i>Pimephales promelas</i> )<br>LC50: 600 - 1000mg/L (96h, <i>Lepomis macrochirus</i> ) | EC50: =55mg/L (48h, <i>Daphnia magna</i> ) |

**Terrestrial ecotoxicity** There is no data for this product.

**Persistence and degradability** No information available.

### Bioaccumulative potential

#### Bioaccumulation

##### Component Information

| Chemical name  | Partition coefficient |
|----------------|-----------------------|
| Diethanolamine | -2.46                 |

### Mobility in soil

**Mobility** No information available.

### Other adverse effects

No information available.

## Section 13: Disposal considerations

### Disposal methods

**Waste from residues/unused products** Dispose of product in packaging in a way that is consistent with the EPA Consolidation 30 April 2021 of the Hazardous Substances (Disposal) Notice 2017 and the Act. Treat the substance using a method that changes the characteristics or composition of the substance so that the substance is no longer a hazardous substance; or export the substance from New Zealand as waste.

Substances which are hazardous to human health or corrosive to metals – may be discharged into the environment if a tolerable exposure limit has been set for the substance (or a component of that substance); and the discharge does not, after reasonable mixing, result in the concentration of the substance in an environmental medium exceeding the tolerable exposure limit. If there is no tolerable exposure limit for the substance, then it may only be discharged into the environment if the substance is very rapidly converted to substances that are not hazardous substances.

Dispose of in accordance with local regulations.

Dispose of waste in accordance with environmental legislation.

#### Contaminated packaging

For packages that have been in direct contact with hazardous substances, the person must ensure that the package is rendered incapable of containing any substance. It must be disposed of in a manner that is consistent with the requirements for disposal of the substance that it contained, taking into account the material the package is manufactured from.

Packages may only be reused or recycled if:

- the substance has a physical hazard other than corrosive to metal, and has been treated to remove any residual contents of the hazardous substance;
- or for substances that have a health or environmental hazard, or corrosive to metal, the contents of the residue in the package are below the threshold for the substance to be classified as hazardous in the Hazardous Substances (Hazard Classification) Notice 2020.

### Section 14: Transport information

#### IATA

Not regulated

#### IMDG

Not regulated

#### Transport in bulk according to Annex II of MARPOL and the IBC Code

No information available

#### Special precautions for user

Please refer to the applicable dangerous goods regulations for additional information

### Section 15: Regulatory information

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

##### National regulations

EPA New Zealand HSNO approval code or group standard To be determined

##### National regulations

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

##### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information

Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information

Controlled substance licenses are required to possess certain explosives, vertebrate toxic agents and fumigants. See Part 7 of the Health and Safety at Work Regulation 2017 for more information

#### International Regulations



**The Montreal Protocol on Substances that Deplete the Ozone Layer** Not applicable

**The Stockholm Convention on Persistent Organic Pollutants** Not applicable

**The Rotterdam Convention** Not applicable

#### **International Inventories**

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>NZIoC</b>         | Contact supplier for inventory compliance status. |
| <b>TSCA</b>          | Contact supplier for inventory compliance status. |
| <b>DSL/NDL</b>       | Contact supplier for inventory compliance status. |
| <b>EINECS/ELINCS</b> | Contact supplier for inventory compliance status. |
| <b>ENCS</b>          | Contact supplier for inventory compliance status. |
| <b>IECSC</b>         | Contact supplier for inventory compliance status. |
| <b>KECL</b>          | Contact supplier for inventory compliance status. |
| <b>PICCS</b>         | Contact supplier for inventory compliance status. |
| <b>AIIC</b>          | Contact supplier for inventory compliance status. |

#### **Legend:**

**NZIoC** - New Zealand Inventory of Chemicals  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**EINECS/ELINCS** - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances  
**ENCS** - Japan Existing and New Chemical Substances  
**IECSC** - China Inventory of Existing Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**AICS** - Australian Inventory of Chemical Substances

### **Section 16: Other information**

**Revision date** 03-Sep-2024

**Revision Note** Significant changes throughout SDS. Review all sections.

#### **Key or legend to abbreviations and acronyms used in the safety data sheet**

**Legend** Section 8: **EXPOSURE CONTROLS/PERSONAL PROTECTION**

|                |                             |             |                                  |
|----------------|-----------------------------|-------------|----------------------------------|
| <b>TWA</b>     | TWA (time-weighted average) | <b>STEL</b> | STEL (Short Term Exposure Limit) |
| <b>Ceiling</b> | Maximum limit value         | <b>Sk*</b>  | Skin designation                 |
| <b>C</b>       | Carcinogen                  |             |                                  |

#### **Key literature references and sources for data used to compile the SDS**

Agency for Toxic Substances and Disease Registry (ATSDR)  
 U.S. Environmental Protection Agency ChemView Database  
 European Food Safety Authority (EFSA)  
 Environmental Protection Agency  
 Acute Exposure Guideline Level(s) (AELG(s))  
 U.S. Environmental Protection Agency Federal Insecticide, Fungicide, and Rodenticide Act  
 U.S. Environmental Protection Agency High Production Volume Chemicals  
 Food Research Journal  
 Hazardous Substance Database  
 International Uniform Chemical Information Database (IUCLID)  
 National Institute of Technology and Evaluation (NITE)  
 Australian National Industrial Chemicals Notification and Assessment Scheme (NICNAS)  
 NIOSH (National Institute for Occupational Safety and Health)  
 National Library of Medicine's ChemID Plus (NLM CIP)  
 National Library of Medicine's PubMed database (NLM PUBMED)  
 U.S. National Toxicology Program (NTP)  
 New Zealand's Chemical Classification and Information Database (CCID)  
 Organisation for Economic Co-operation and Development Environment, Health, and Safety Publications  
 Organisation for Economic Co-operation and Development High Production Volume Chemicals Programme  
 Organisation for Economic Co-operation and Development Screening Information Data Set  
 World Health Organization

#### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**